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SUMMARY

Robotics Engineering student with a strong passion for ROS2, Computer Vision, and Autonomous Systems. Proven logical problem-solving skills through the development of a 5-DOF robotic arm (WhitePost). Experienced in Python/C++

EDUCATION

Bachelor of Engineering (B.Eng.) in Robotics Engineering and AI Jun 2022 - Mar 2026

Relevant Coursework : Robot Design Essential, Computer Vision, Robotic Engineering, System Analysis and Control, Embedded Systems, ROS2 , AI for Robotics

Chiang Mai University
GPAX : 3.05

EXPERIENCE & INTERNSHIP

Jinpao Automation Contest 2023 Nov 2022 - Feb 2023

- Designed the robot's mechanical chassis and structural components for optimal stability.
- Collaborated with a multidisciplinary team to implement an automated system for researcher tasks.

IEEE Region 10 Robotics Competition 2024 Mar 2024 - May 2024

- Qualified as Stage 2 Finalist in a international robotics competition.
- Developed an autonomous Floating Robot (ASV) for collect environmental water quality sampling.

TESR Co.,Ltd. Internship Apr 2025 - May 2025

- Developed a high-payload Autonomous Mobile Robot (AMR) capable of transporting up to 300 kg.
- Collaborated on hardware-software integration by interfacing ROS2 with ESP32 via Serial communication to control Hub Servo Motor drivers.

PROJECT

Turtlebot3 : SLAM Navigation Jan 2024 - Feb 2024

- Deployed SLAM (Lidar-based) and Nav2 stack for autonomous environment mapping and path planning
- Visualized and debugged transform trees (TF) and laser scan data via Rviz2 and Gazebo.

Orange Robot : a customized mobile robot platform Feb 2024 - Mar 2024

- Developed a mobile robot platform integrated with a 5-DOF robotic arm for a Teleoperation system.
- Simulated robot kinematics and navigation in Gazebo using ROS2 .
- Manually navigated the robot to target locations and operated the robotic arm to complete precise orange box pick-up and drop-off tasks.

Service Robot for Dental Clinic CMU Jun 2024 - Oct 2024

- Designed and developed a Service Robot prototype for dental clinic environments, integrated with a 5-DOF robotic arm.
- Developed a prototype mobile robot with an arm to manually transport dental trays from the sterilization room to treatment rooms.

White Post : a robot document delivery Jun 2025 - Mar 2026

- Designed the custom robot chassis to support a 5-DOF arm and integrated a tracked and wheel drive system for multi-terrain mobility.
- Developed full control software for a 5-DOF robotic arm, implementing Inverse Kinematics (IK) and Forward Kinematics (FK) for precise button lift handling.
- Developed and deployed a custom YOLOv8 model integrated with OpenCV to identify and track elevator buttons and monitor elevator door status (Open/Closed).

SKILLS

- **Programming Language** : Python, C++, C, Lua
- **Robotics Technical Skills** :
  - **Hardware:** TurtleBot3 Burger/Waffle, 2D LiDAR, OpenCR Controller, Raspberry Pi 5/Zero 2W, ESP32, Arduino Uno
  - **Navigation & SLAM:** SLAM (Lidar-based), Nav2, Path Planning (RRT, RRT*, A*)
  - **Frameworks & Tools:** ROS2 (Foxy, Humble), Rviz2, Gazebo Classic
- **Computer Vision & AI:** OpenCV, YOLOv8, Image Processing, Roboflow, Large Language Models (ChatGPT/Gemini API Integration),
- **Industrial Automation:** Siemens PLC , HMI Design, TIA Portal v16, Ladder Logic
- **Language:** Thai (Native) , English (Basic)
- **Tools & Development:** Git/GitHub, Linux/Ubuntu, Visual Studio Code, SolidWorks, EasyEDA, Arduino IDE
- **Soft Skills:** Logical Problem Solving , Technical Communication, Team Collaboration, Rapid Self-Learning